CSCI 5526

Unsupervised Learning in Healthcare II

Announcements

- MP2 will be released today!

Student Questions

You mentioned using PCA on brain PET images in class. The three important components explained only 9%, 8%, and 4% of the total variance in the data. So, could PCA discard some low-variance features that carry the most clinical information? How could we avoid this?

-- Yes, you must always inspect the components before throwing them away.

-- Sometimes you can have a criterion that is more clinically guided to determine which components to keep.

What are the main limitations of using PCA within clinical data?

-- Linearity Assumption

-- Vulnerability to Outliers

-- Loss of Interpretability

-- Variance vs. Importance Mismatch

Student Questions

How do we know the correct metrics for our use case? In healthcare what metrics matter the most?

Recall / Sensitivity (True Positive Rate): This is often the top priority. Missing a disease like cancer or sepsis (a false negative) can lead to severe harm or death.

Specificity (True Negative Rate): It matters when you want to avoid scaring healthy patients or burdening clinics with unnecessary follow-up tests.

Precision (Positive Predictive Value): It is vital when a false positive triggers invasive treatments, extreme anxiety, or a waste of limited hospital capacity.

AUPRC (Area Under the Precision-Recall Curve): In medicine, many conditions are rare. AUPRC is more reliable than standard AUC-ROC for imbalanced datasets.

For imbalanced data, what methods can we use during model training to improve sensitivity, even if it leads to a lower accuracy?

-- Adjust class weights in loss functions

-- Use sampling to increase minority samples or decrease majority samples

-- Use focal loss to prioritize harder minority samples

Student Questions

Are pet scans expansive? If they can detect early Alzheimer's and similar, why aren't they part of routine check for 60+ people

-- Medicare covers FDG-PET scans for Alzheimer's disease only when used for a differential diagnosis to distinguish between Alzheimer's disease and fronto-temporal dementia (FTD).

-- It is NOT covered by insurance for routine screening, since it is still in research phase.

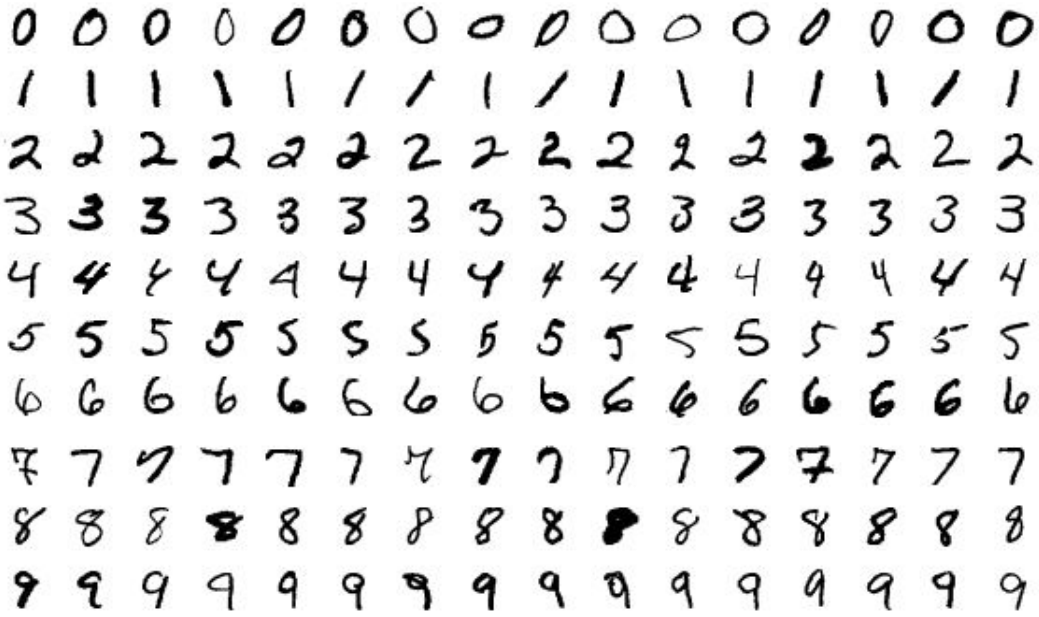
Outline

- Dimensionality reduction for visualization
  - tSNE – t-distributed Stochastic Neighborhood Embedding
- Clustering goals
- Examples in healthcare
- Defining clusters
  - Hard vs soft clustering
  - Distance measures
- Algorithms
  - Hierarchical clustering
  - DBSCAN
- Evaluating clustering results

Non-linear Dimensionality Reduction for Visualization

- Examples: t-SNE, UMAP
- Primary (and only) purpose: visualization
- Transforming new data is not the goal.

Example: MNIST dataset

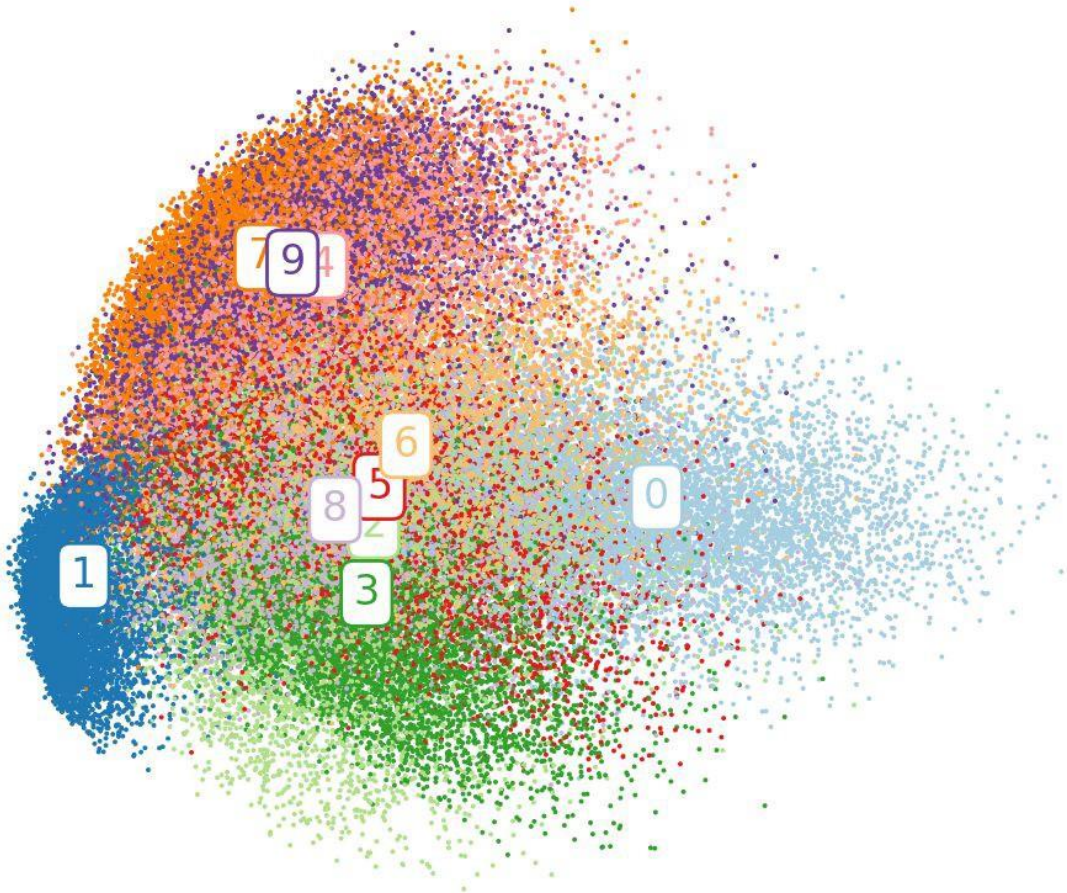


n = 70,000

28×28 images = 784 pixels

Slide credit: Dmitry Kobak (Tubingen)

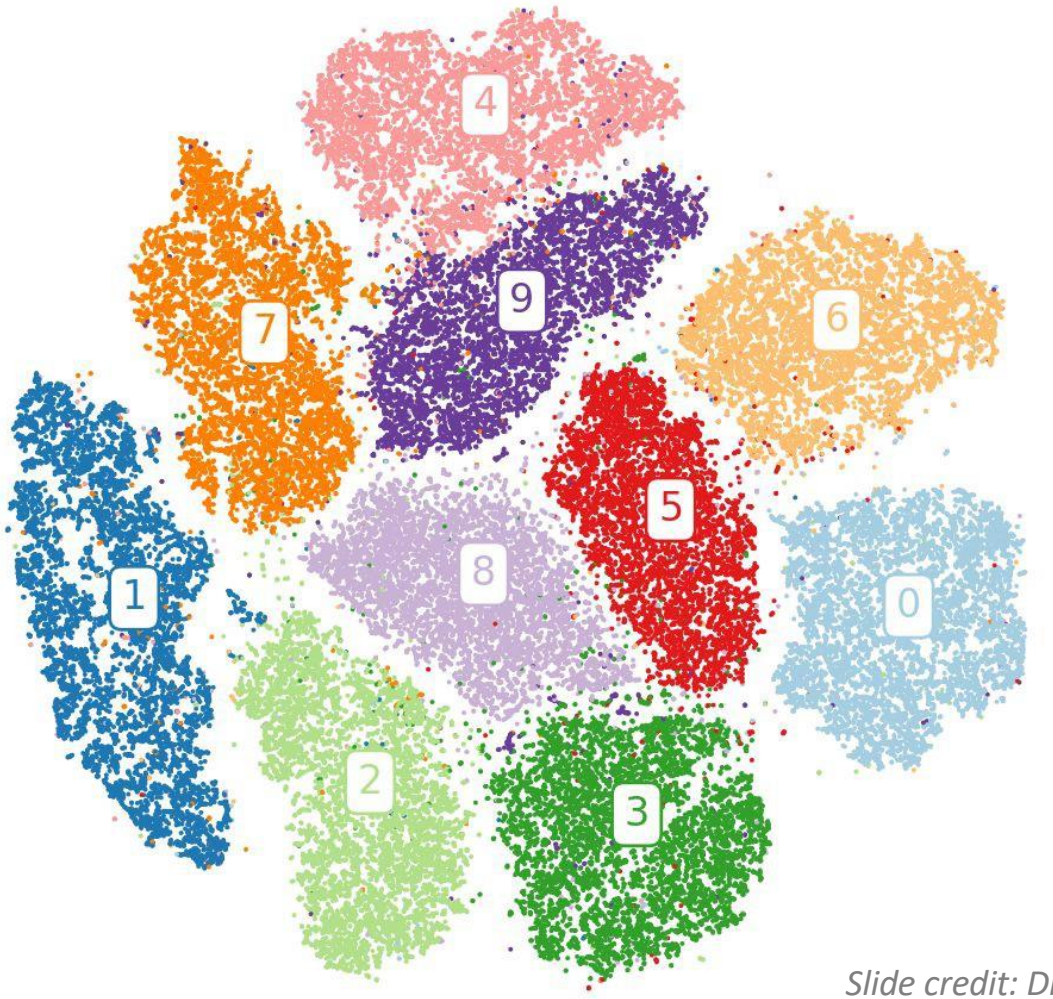
MNIST dataset: PCA





Slide credit: Dmitry Kobak (Tubingen)

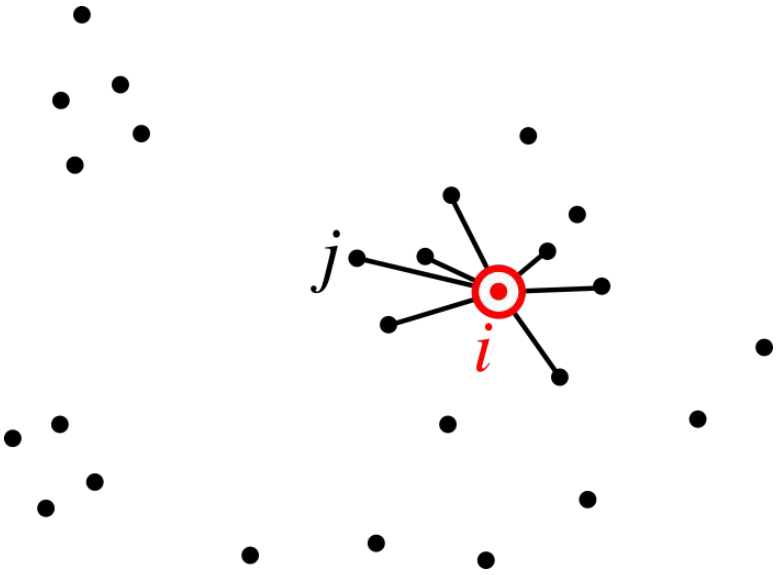
MNIST dataset: t-SNE



Slide credit: Dmitry Kobak (Tubingen)

Stochastic neighbor embedding

- Loss function — Kullback-Leibler divergence between pairwise similarities (affinities) in the high-dimensional and in the low-dimensional spaces.
- Similarities are defined such that they sum to 1.



$$\mathcal{L} = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

High price for putting close neighbours far away.

Slide credit: Dmitry Kobak (Tubingen)

Stochastic neighbor embedding

- High-dimensional similarities:

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma_i^2)}{\sum_{k\neq i}\exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2/2\sigma_i^2)}$$

- Kernel width is adaptively chosen to achieve the desired perplexity (default 30).

- Then symmetrize and normalize to sum to one:

$$p_{ij} = \frac{p_{i|j} + p_{j|i}}{2n}$$

Slide credit: Dmitry Kobak (Tubingen)

Stochastic neighbor embedding

Low-dimensional similarities:

$$q_{ij} = \frac{w_{ij}}{Z}, \quad w_{ij} = k(\|\mathbf{y}_i - \mathbf{y}_j\|), \quad Z = \sum_{k \neq l} w_{kl}$$

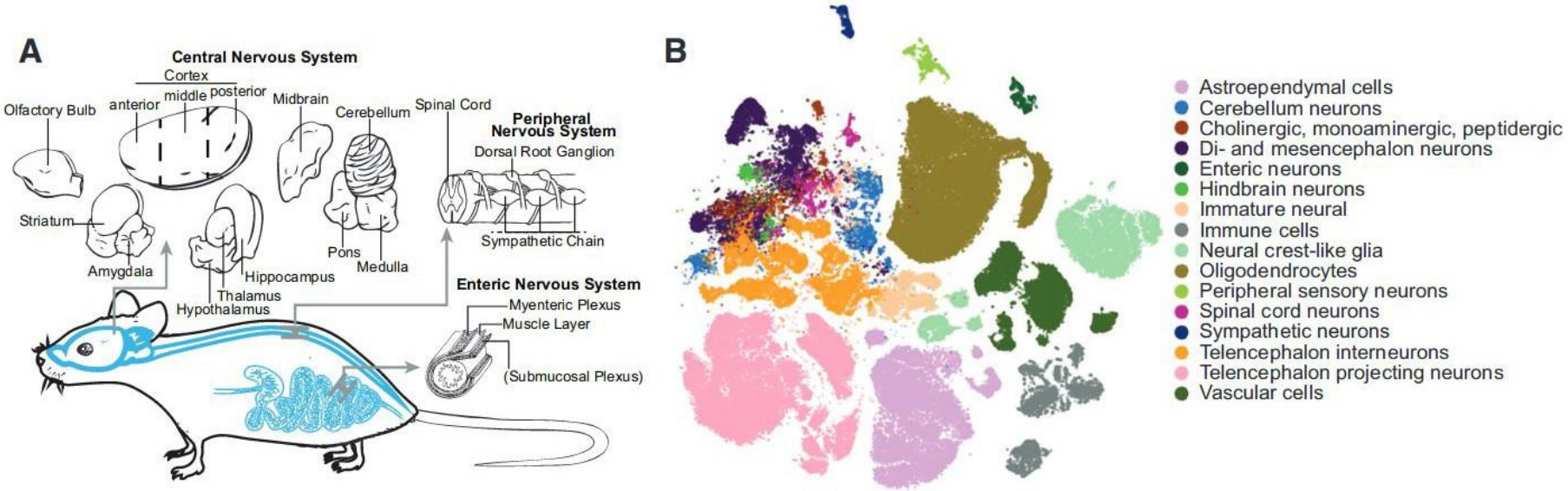
Similarity kernel in t-SNE (t-distribution):

$$k(d) = 1/(1 + d^2)$$

Slide credit: Dmitry Kobak (Tubingen)

t-SNE Examples

Single-cell transcriptomics (single-cell RNA sequencing): samples are cells, features are genes.

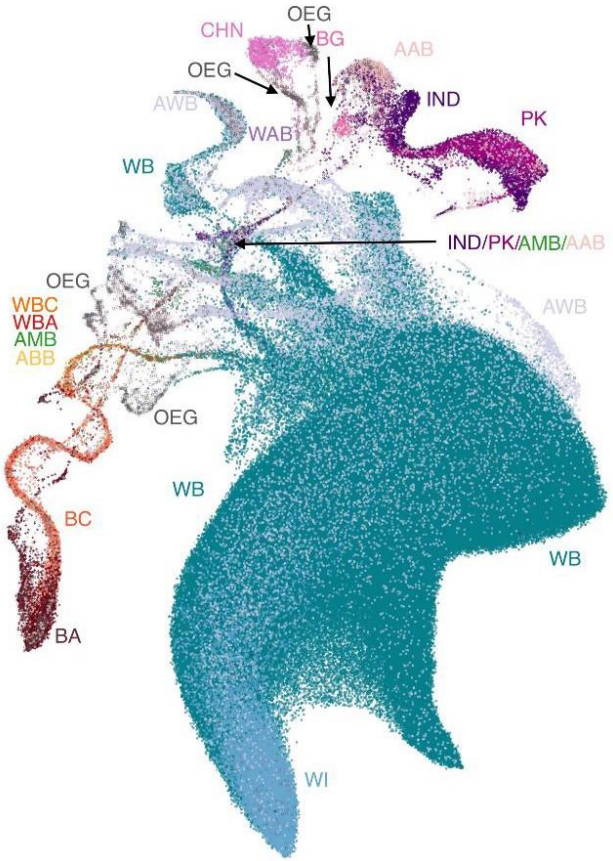


Zeisel et al. (2018)
n ≈ 500,000

Slide credit: Dmitry Kobak (Tubingen)

t-SNE Examples

Population genomics: samples are people, features are single-nucleotide polymorphism (SNPs).



Diaz-Papkovich et al. (2019)
n ≈ 500,000

Slide credit: Dmitry Kobak (Tubingen)

Clustering

- Clustering refers to a very broad set of techniques for finding subgroups, or ***clusters***, in a data set.

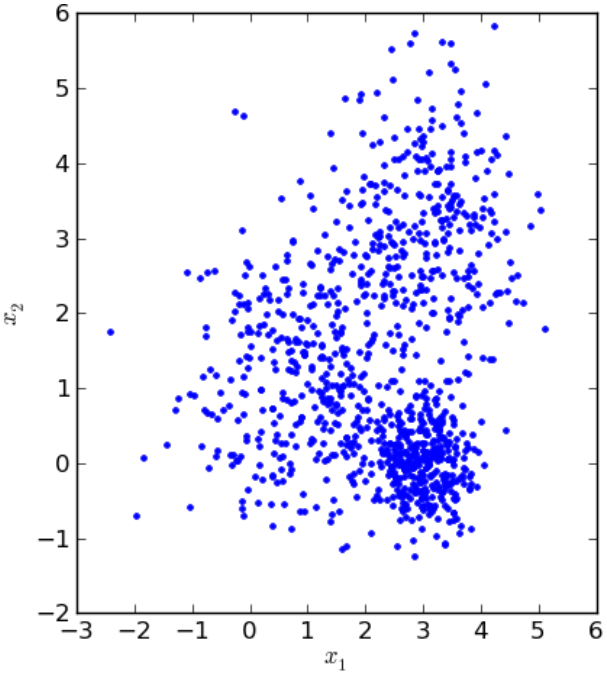
- We seek a partition of the data into distinct groups so that the observations within each group are similar to each other.

Clustering in Healthcare

- Segmentation (without ground truth data)
  - Which part looks different from the rest?
- Disease subtype discovery
  - Same symptoms but different underlying causes
  - Same underlying cause but different symptoms or trajectories
- Treatment subgroups
  - Subgroups based on comorbidities, family history, genetics
  - How different people respond to the same drug/treatment
- To guide labeling
  - Cluster data into groups and label the groups.

Clustering Questions

- How many clusters are there?
- What does it mean for two or more observations to be similar or different?
  - This is often a domain-specific consideration that must be made based on knowledge of the data being studied.
- How to define cluster boundaries?
- How to evaluate clustering results?



Distance Measures

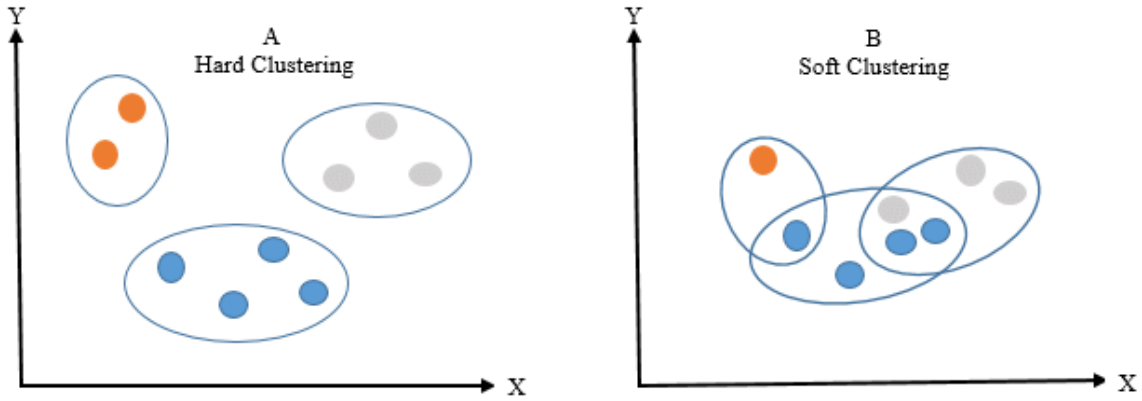
- Measure how 'similar' two samples are.

Euclidean	$d(x,y) = \sqrt{\sum (x_i - y_i)^2}$
Squared Euclidean	$d(x,y) = \sum (x_i - y_i)^2$
Manhattan	$d(x,y) = \sum \lvert (x_i - y_i) \rvert$
Canberra	$d(x,y) = \sum \frac{\lvert x_i - y_i \rvert}{\lvert x_i + y_i \rvert}$
Chebychev	$d(x,y) = \max(\lvert x_i - y_i \rvert)$
Bray Curtis	$d(x,y) = \frac{\sum \lvert x_i - y_i \rvert}{\sum x_i + y_i}$
Cosine Correlation	$d(x,y) = \frac{\sum (x_i y_i)}{\sqrt{\sum (x_i)^2 \sum (y_i)^2}}$
Pearson Correlation	$d(x,y) = \frac{\sum (x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum (y_i - \overline{y})^2} \sqrt{\sum (y_i - \overline{y})^2}}$

Boundaries: Hard vs Soft Clustering

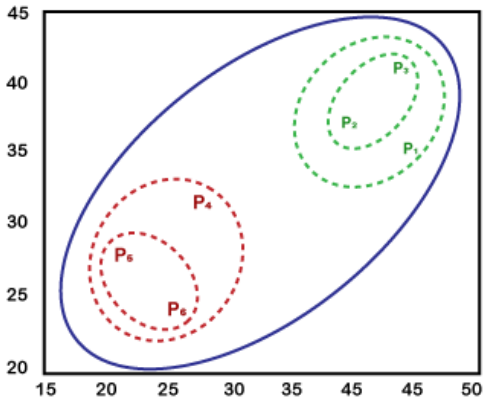
- Hard clustering
  - Each sample is assigned to only one cluster.
  - Clusters do not overlap.

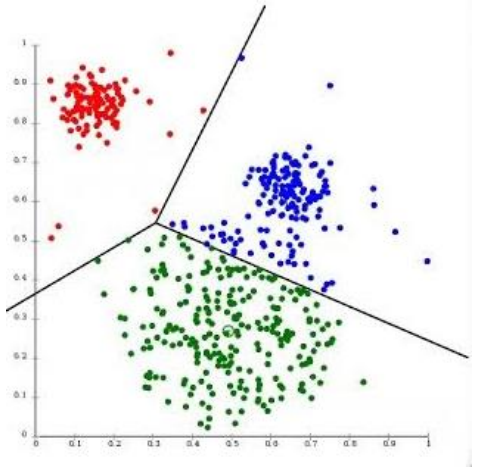
- Soft clustering
  - Every sample is assigned a probability for a cluster.
  - Clusters may overlap.

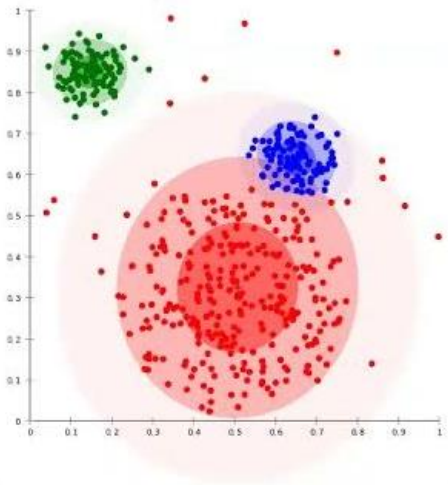


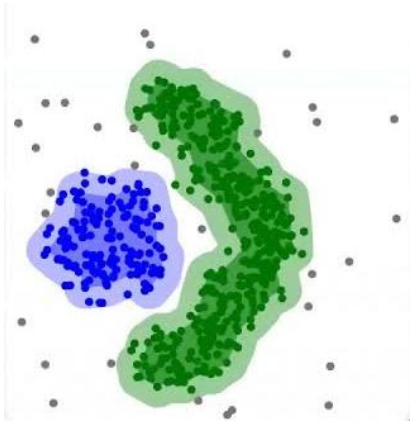
Clustering Algorithms

- Centroid-based
  - k-means
- Hierarchical
  - Agglomerative clustering
- Density-based
  - DBSCAN
- Distribution-based
  - Gaussian mixture model

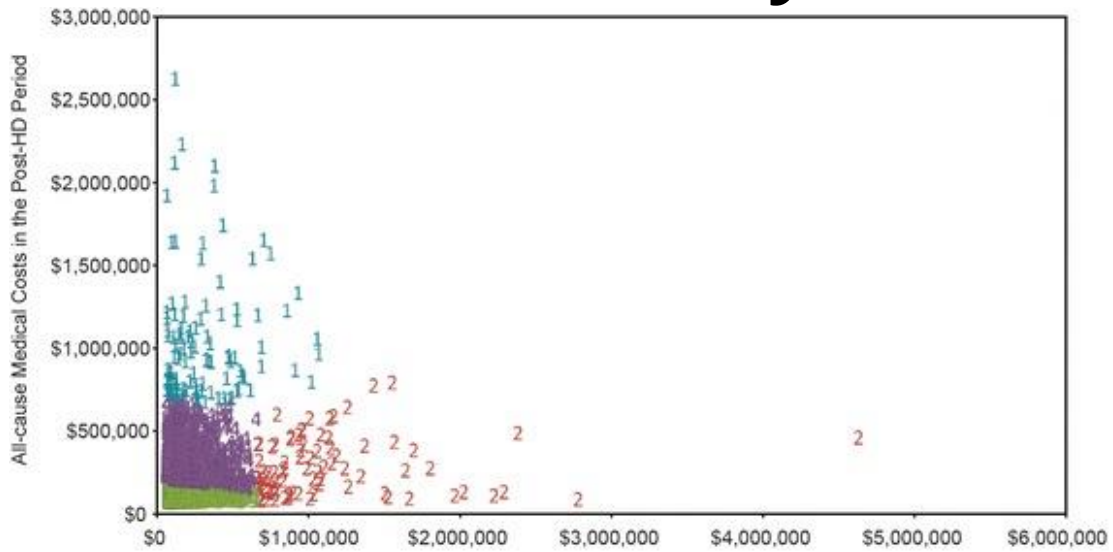


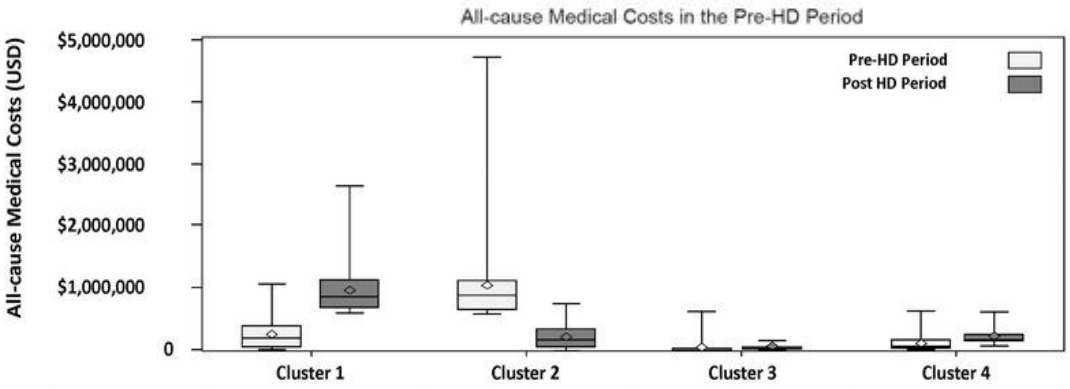






Medical Cost of Hemodialysis (HD) in Chronic Kidney Disease Patients





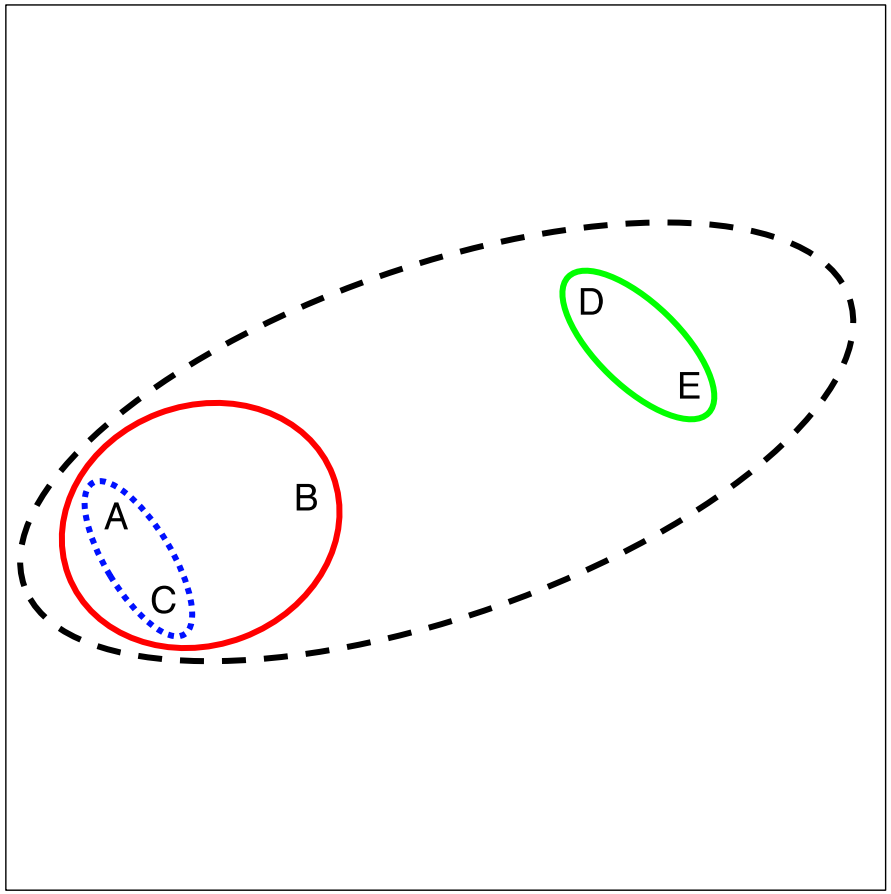
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4776444/

- Claims data from >18,000 patients.
- Pre-HD and post-HD costs within 12-months were clustered.
- Identified 4 clusters:
  - Average->High
  - Very high->High
  - Average->Average
  - High->High
- Comorbidities were the main drivers of costs going up from pre-HD to post-HD.

Hierarchical Clustering

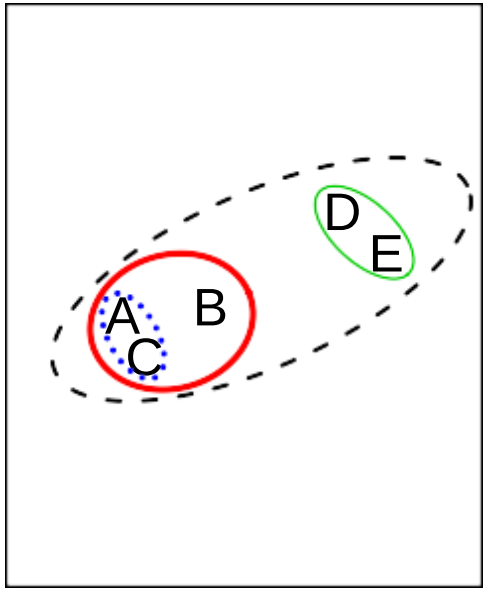
- Works by creating a hierarchy based on similarity.
- Two different types:
  - Agglomerative: Initially, each sample is in its own cluster. The clusters are then merged step by step until a single cluster remains.
  - Divisive: Initially, all objects are in a single cluster. Then the division process is performed step by step until each sample forms a different cluster.
- Will describe agglomerative, more popular of the two.

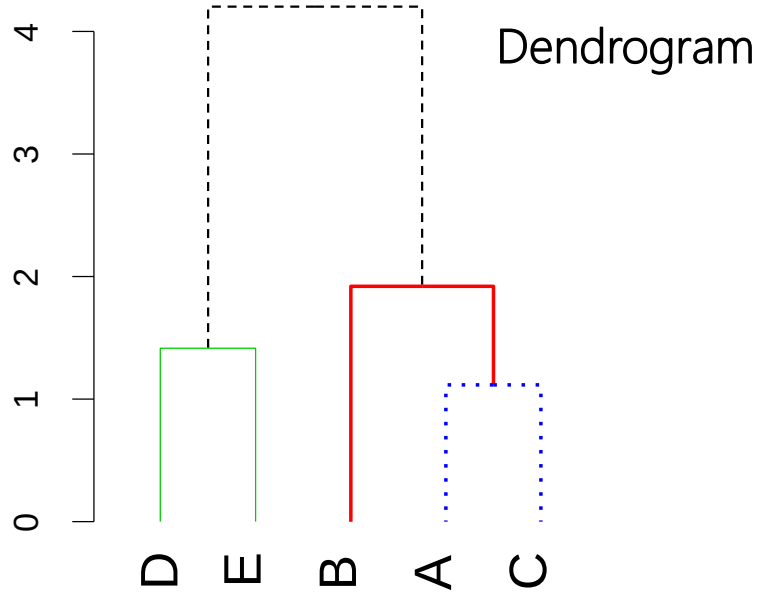
How it works



Hierarchical Clustering Algorithm

- The approach in words:
- Start with each point in its own cluster.
- Identify the closest two clusters and merge them.
- Repeat.
- Ends when all points are in a single cluster.





Grouping Clusters - Distances

- **Single Linkage**

  $D(c_1,c_2) = \min D(x_1,x_2)$

  Minimum distance or distance between closest elements in clusters



- **Complete Linkage**

  $D(c_1,c_2) = \max D(x_1,x_2)$

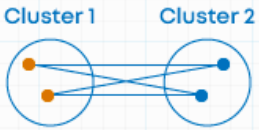
  Maximum distance between elements in clusters



- **Average Linkage**

  $D(c_1,c_2) = \frac{1}{|c_1|}\frac{1}{|c_2|}\Sigma\Sigma D(x_1,x_2)$

  Average of the distances of all pairs

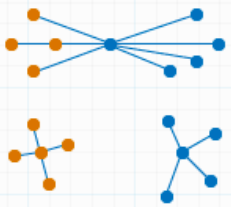


- **Centroid Method**

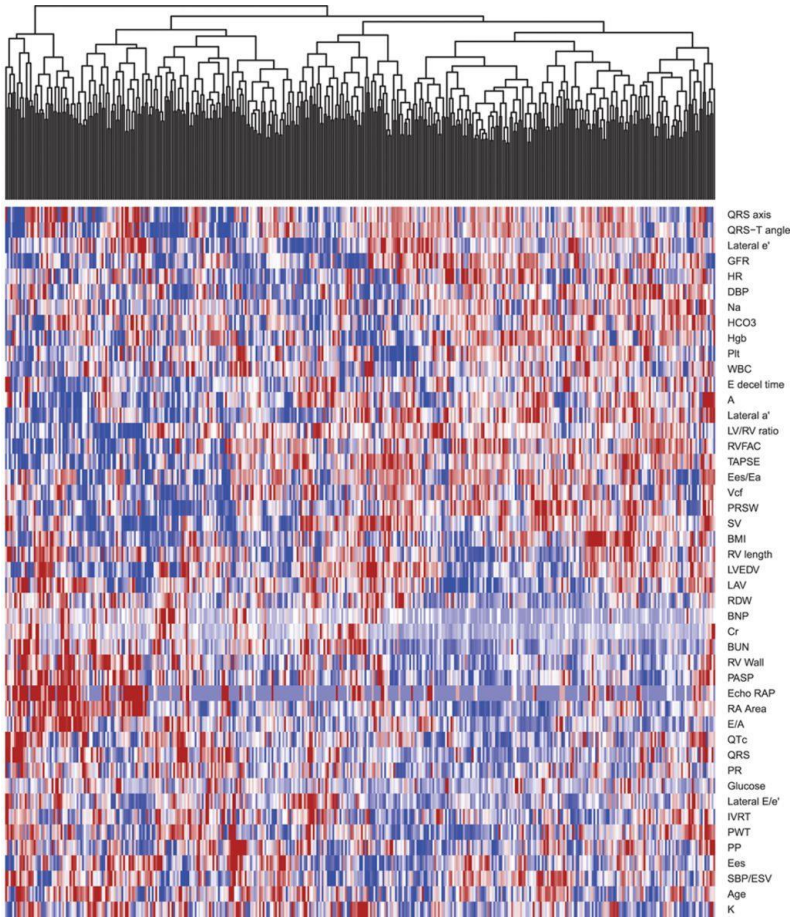
  Combining clusters with minimum distance between the centroids of the two clusters

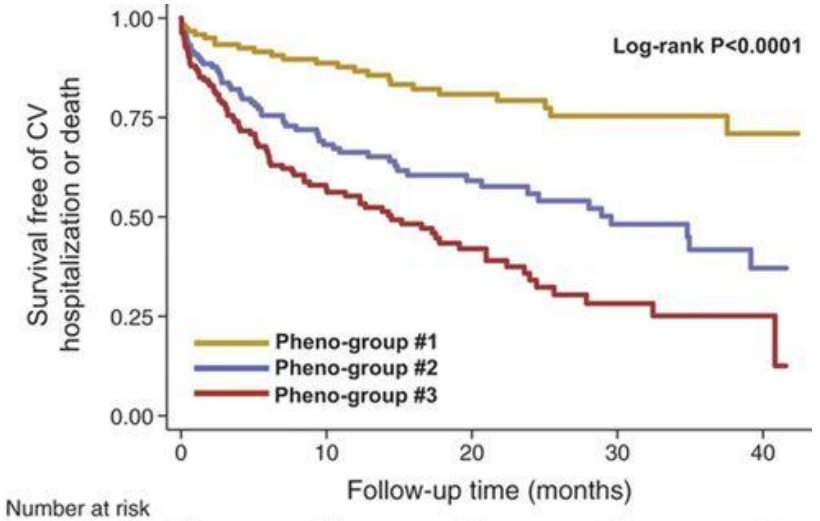


- **Ward's Method**
  - Combining clusters where increase in within cluster variance is to the smallest degree.
  - Objective is to minimize the total within cluster vairance



Subtype Discovery in Heart Failure





Source: https://doi.org/10.1161/CIRCULATIONAHA.114.010637

Hierarchical Clustering: Pros and Cons

- Pros
  - k need not be specified. We can obtain the optimal number of clusters from the model itself, human intervention not required.
  - Dendrograms help us in clear visualization, which is practical and easy to understand.

- Cons
  - Not suitable for large datasets due to high time and space complexity.
  - All the approaches to calculate the similarity between clusters has their own disadvantages.

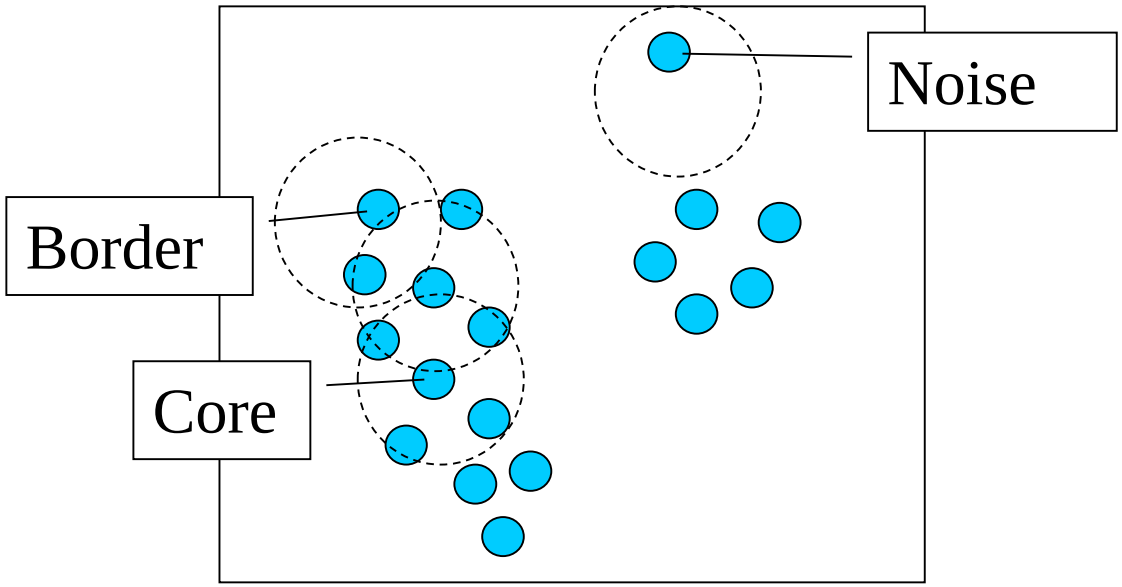
DBSCAN

- DBSCAN: **d**ensity-**b**ased **s**patial **c**lustering of **a**pplications with **n**oise

- Major features:
  - Discover clusters of arbitrary shape
  - Handle noise

- Two parameters:
  - *Eps*: radius of the neighborhood
  - *MinPts*: minimum number of points in an Eps-neighborhood

DBSCAN: Core, Border, and Noise Points



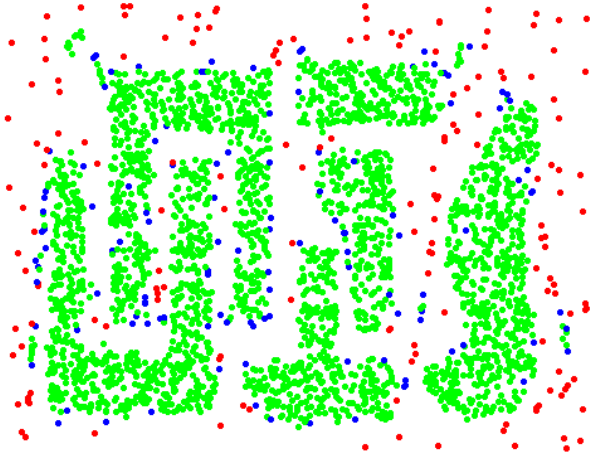
ε = 1unit, MinPts = 5

- Density = number of point within a specified radius (Eps)
- A point is a core point if it has more than a specified number of points (MinPts) within Eps
  - These are points that are at the interior of a cluster
- A border point has fewer than MinPts within Eps, but is in the neighborhood of a core point
- A noise point is any point that is not a core point or a border point.

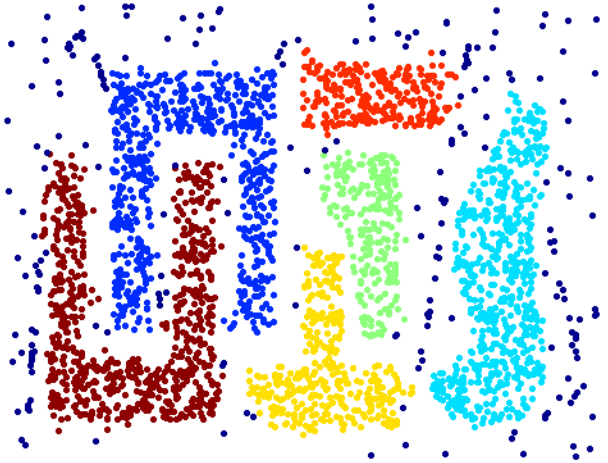
Example: DBSCAN



Original Points



Point types: core, border and noise



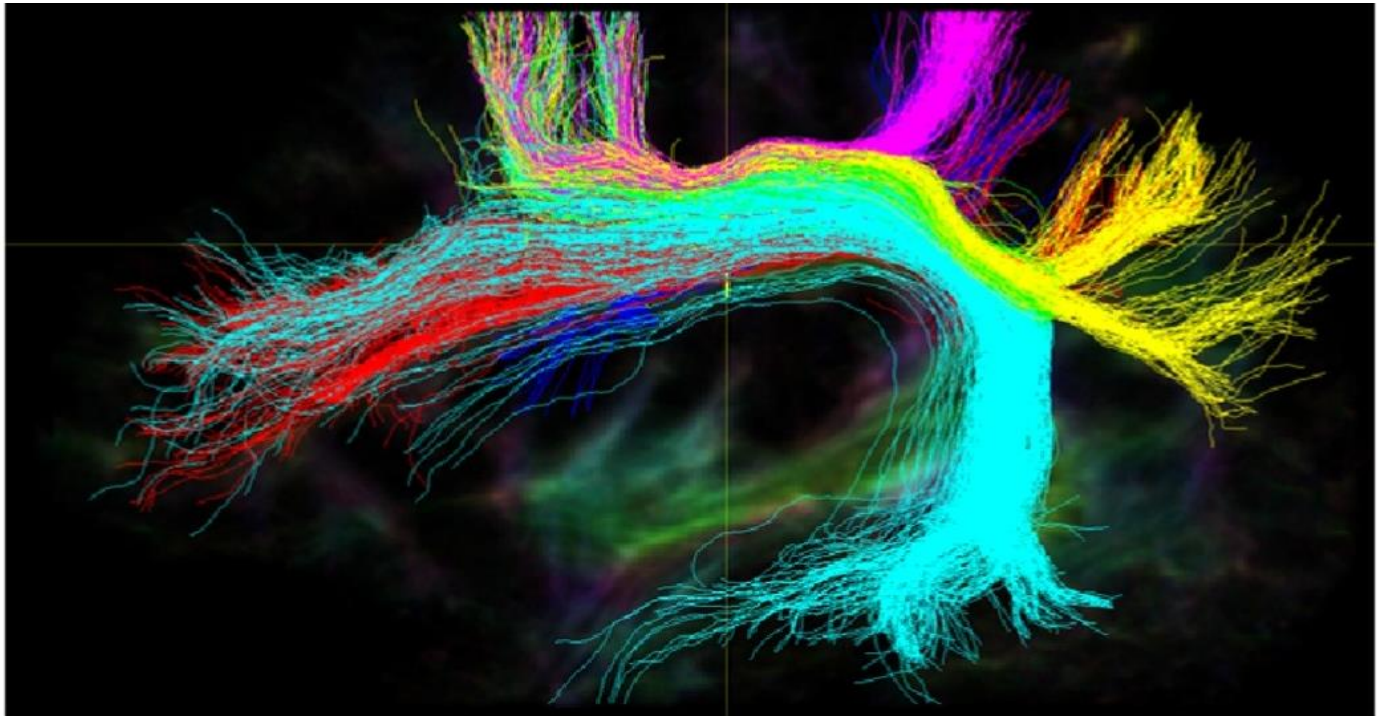
Clusters

DBSCAN: Pros & Cons

- Pros
  - *k* need not be specified.
  - Resistant to outliers.
  - Can handle clusters of different shapes and sizes.

- Cons
  - Will not handle different densities.
  - Need hyper-parameter tuning (Eps, MinPts). Can be difficult for high-dimensional data.
  - No easy visualizations like dendrograms.

Clustering MR-Tractography Data



Source: https://doi.org/10.1016/j.jneumeth.2019.108361

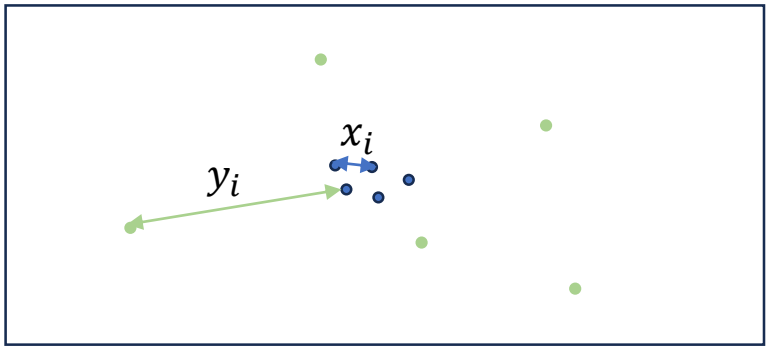
Evaluating Clustering

Slides adapted from Jiawei Han, Micheline Kamber, and Jian Pei

Assessing Clustering Tendency

- Assess if non-random structure exists in the data by measuring the probability that the data is generated by a uniform data distribution
- Test spatial randomness by statistic test: Hopkins Static
  - The Hopkins Statistic:

$$H = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} y_i}$$



  - If there is no structure, H is close to 0.5. If there is structure, H is close to 1.

  - In Python: https://pypi.org/project/pyclustertend/

Determining the Number of Clusters

- Elbow method
  - Use the turning point in the curve of sum of within cluster variance with respect to the # of clusters (k).
- Cross validation method
  - Divide a given data set into m parts.
  - Use m-1 parts to obtain a clustering model.
  - Use the remaining part to test the quality of the clustering.
    - E.g., for each point in the test set, find the closest centroid, and use the sum of squared distance between all points in the test set and the closest centroids to measure how well the model fits the test set.
  - Repeat it m times, compare the overall quality measure with respect to different k's, and find # of clusters that fits the data the best.

Measuring Clustering Quality

- Two methods: extrinsic and intrinsic
- Extrinsic: supervised, i.e., the ground truth is available.
  - Compare a clustering against the ground truth using certain clustering quality measure.
- Intrinsic: unsupervised, i.e., the ground truth is unavailable.
  - Evaluate the goodness of a clustering by considering how well the clusters are separated, and how compact the clusters are.

Extrinsic Methods

- Clustering quality measure: Q(C, Cg), for a clustering C given the ground truth Cg.

- Q is good if it satisfies the following criteria.
  - Cluster homogeneity: A clustering result satisfies homogeneity if all of its clusters contain data points which are members of a single class only.

  - Cluster completeness: A clustering result satisfies completeness if all the data points that are members of a given class are elements of the same cluster.

- v-measure (beta is a weight specified in [0,1])

```
v = (1 + beta) * homogeneity * completeness / (beta * homogeneity + completeness)
```

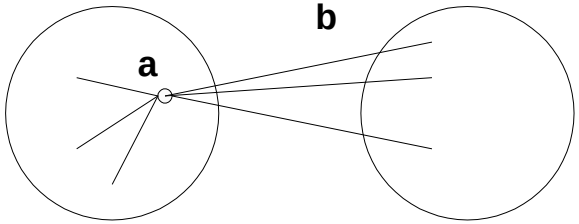
Intrinsic Measures: Silhouette Coefficient

- Silhouette Coefficient: a measure based on intra-cluster distances and inter-cluster distances.

- For an individual point, i
  - Calculate a = average distance of i to the points in its cluster
  - Calculate b = min (average distance of i to points in another cluster)
  - The silhouette coefficient for a point is then given by

$$s = \frac{b - a}{b}$$

  - Typically, between 0 and 1.
  - The closer to 1 the better.



- Can calculate the average Silhouette width for a cluster or all clusters.

Exit pass

https://z.umn.edu/ml4h_exitpass

